

ImageEn Paint Engine (Bundle 3.2)

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Class TImageEnPaintEngineManager

TComponent <-- TImageEnPaintEngineManager

Declared in Unit NWSComps_IEPaintengine_Manager

Description:

This is the user interface class by which the user can configure, start and stop the Paint Engine. It is a visual component that can be placed on a form.

Properties

property EditedRect: Trect read-only

Gives the coordinates of the lastly edited area.

property IsRunning: boolean read-only

Checks whether the Paint-engine is running.

property Params_Cloning

Parameters to control the behavior of PE in the Cloning Mode:

→ ShowBrush:boolean → when true the PE will display the shape of the cursor at the clone picture source point

→ SrcImageEn: TimageenView → the source ImageEnView component from which to clone

→ srcPoint:TPoint → the source point

→ tgtPoint:TPoint → the target point

property Params_Color

Parameters to control the behavior of PE in the Color Mode:

→ Color:Tcolor → the color to be used when painting in color mode

property Params_History

Parameters to control the behavior of PE in the History Mode:

→ HistoryBitmap: TIEBitmap → the image that will be restored by painting when the PE is in history mode

property Params_Paint

Parameters to control the general behavior of PE:

→ BlendMode: Tblendmode → the current blend mode to be used during painting.

→ BrushBitmap:TIEBitmap → the picture to be used as a brush. If not assigned the standard round brush will be used

→ ContinuousPainting: boolean → when true the PE will never stop painting even if the user does not move the cursor. If false the painting will stop when the user does not change the cursor position.

- Dithering:integer (0..100) → the amount of dithering to apply to the default round brush
- EnableAutoScroll:boolean → When true, while painting if the cursor goes beyond the display rectangle, the PE will scroll the display.
- EnableEditAlphaChannel:boolean → when true the alpha channel of the current layer will be also edited while painting
- Feather:integer (0..100) → the amount of smoothing of the default round brush
- Gradation:integer (0..100) → the gradation of the smoothing of the default round brush
- LargeSteps: boolean → when true the pe will not draw a line between two consecutive brush positions
- Mode: tPeMode → the mode of painting. Can be pemColor, pemRetouch, pemHistory, pemClone, pemTexture, pemDeformations, pemEraseLayer, pemRestoreLayer
- Precision: integer (0..100) → the amount of precision, the higher the more cursor positions will be elaborated.
- Radius: integer → the radius in pixels of the brush
- RotateBrush: boolean → when true the PE will rotate the brush according to the angle of movement.
- Selective: boolean → when true the painting will affect only those parts of the picture whose color matches the color of the region of the picture in which the painting has started
- SelTolerance: integer (0..255) → the tolerance for the selective painting algorithm
- SessionMode: tPeSessionMode
 - sm_IgnoreSessionMemory: in this mode the engine always build up strokes
 - sm_KeepSessionMemory_UntilNext: in this mode during the current painting session strokes are not built up, but in the next painting session, strokes will be build up on top of the previous painting
 - sm_KeepSessionMemory_UntilManualReset: in this mode strokes are not built up until you reset the session manually using the Session_Reset method.
- ShowBrushShape: boolean → when true the PE will display the shape of the brush at the cursor when painting
- SprayEffect: (0-100) the higher the value the more the brush will simulate a spray effect
- Step:integer distance in pixels between two consecutive brush strokes.
- Transparency: integer (0..255) the higher it is the less pressure will be on the painting. Set it to 0 paint at full pressure.

property Params_Retouch

Parameters to control the behavior of PE in the Retouch Mode:

- Amount:integer (0..255) the amount of retouch to be applied
- Kind: TpeRetouchKind:
 - rk_blur: blurs the picture by averaging the colors of the painted area
 - rk_Sharpener: sharpens the edges of the picture in the painted area
 - rk_dodge: brightens (dodge mode) the picture in the painted area
 - rk_burn: darkens (burn mode) the picture in the painted area rk_Lighten: brightens the picture in the painted area
 - rk_Darken: darkens the picture in the painted area
 - rk_Saturate: increases the saturation in the painted area
 - rk_Desaturate: decreases the saturation in the painted area
 - rk_Custom: use your own custom function to retouch. To define your custom function use the RetouchCustom property
- RetouchCustom: TPERetouchCustomFunction = function(const inputBGRA: TbeBGRAbytearray):

TbeBGRABytearray of object;

property Params_Tablet

Parameters to control the behavior of Graphic Tablet:

- PressureMode → can be pmLinear or pmUnlinear
- PressureSensitive:boolean when true the PE will read the pressure changes from the tablet
- PressureUnlinearStrength: integer (0..100) the higher the more unlinear the sensitivity to the pressure
- PressureVar: integer → pressure variation scale. This influences the way the amount of pressure is accounted
- PressureDynamics: you can specify dynamics to change either the Opacity or the Radius of the brush, or both, using the tablet pressure sensor.

property Params_Texture

Parameters to control the behavior of PE in the Texture Mode:

- TextureBitmap: TIEBitmap → the picture to be used as texture

property Params_Warp

Parameters to control the behavior of PE in the Warp (Deformation) Mode:

- Amount:integer (0..255) the amount of warp effect to be applied
- Kind: TpeDeformationKind:
 - dk_WhirlCW: whirlpool clockwise
 - dk_WhirlACW: whirlpool anticlockwise
 - dk_Bulge: bulge effect
 - dk_Pinch: pinching effect
- Step: integer (0..255) The step by which the amount of warp must increase with time
- StepTime: integer (milliseconds) → the time interval before the amount of the effect is increased

property UpdateMode

Changes the mode of update. The PE can call an imageEn update at each single brush stroke or at the painting of each line. The single brush mode is generally to be preferred, unless you have a picture with a lot of image layers

- umEachStroke: the update happens at each stroke
- umEachLine: the update happens at each line

property UpdatePriority

Set the priority of updating the display of ImageEn. By default the priority is high.

- upHigh: High speed update
- upMedium: Middle speed update
- upLow: Low speed update

With big images you may desire a lower speed of update.

Events

property OnFinishedPainting: TNotifyEvent

Handle this event to perform any duty that must be done after the PE has stopped painting.

property OnPainting: tpeNotifyEvent = procedure(sender: Tobject; var x, y: integer) of object;

Handle this event to perform any duty that must occur while the PE is painting, such as displaying the editing coordinates.

property OnStartedPainting: TNotifyEvent

Handle this event to perform any duty that must be done after the PE has started painting.

property OnSessionEvent: tPeParamsSessionEvent = procedure(ChangedParams: tPeCustomParams; Session: TPeSession) of object;

Only for advanced use. Handle this event to perform any duty that must occur when the state of the painting session has changed

Methods

procedure Session_AddFlag(const theFlagName: string)

For advanced use. You can use flags to attach to the current session some information that you may use to remind you of the status of the session.

procedure Session_RemoveFlag(const theFlagName: string)

For advanced use. Removes a flag previously added

procedure Session_Reset

Resets the memory of the current painting session manually. This is useful only if you use the advanced session mode: sm_KeepSessionMemory_UntilManualReset
Most users will not need this mode.

procedure Init(theImageEn: TImageenview; theMode: tpemode);

Initializes the PE by specifying on which TimageEnView the painting should occur and the mode of painting.

StartCursorNoPaint(theimageen: TImageenview; radius: cardinal);

Call this method to show the cursor even when painting has not started on the specified ImageenView. When you change the radius, you have to call this method again, to reflect the current radius.

procedure StartPainting overload

Commands the PE to start painting. After this command the PE will set itself into running mode and will trigger the StartedPainting Event.

procedure StartPainting(ForceNewSession: boolean); overload

Commands the PE to start painting and specifies whether a new session should be enforced (which is equivalent to call the Session_Reset method). After this command the PE set itself into running mode and will trigger the StartedPainting Event.

procedure StopCursorNoPaint

Call this to stop the display of the cursor on the attached Imagenenview, when not yet painting. It is good practice to call it also on the destroy event of the form.

procedure StopPainting

Commands the PE to stop painting. After this command the PE will finalize the running mode and will trigger the FinishedPainting Event.